

■ DOCFLOW AI

DocFlow AI

AI-Powered Document Intelligence Platform

STREAMLIT

ANTHROPIC API

PYTHON 3

AI / ML

Project Type	AI-Powered Web Application
Platform	Streamlit Dashboard
AI Engine	Claude Sonnet 4 (Anthropic API)
Version	1.0.0
Date	May 24, 2026
Status	Completed ✓

Created by

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1. Executive Summary

DocFlow AI is a cutting-edge, AI-powered document intelligence platform built using Python and Streamlit, integrated with Anthropic's Claude AI model. The platform dramatically reduces the time and effort required to create professional documents, analyze resumes, and generate structured content for academic, professional, and business purposes.

The platform serves students, professionals, researchers, and businesses — enabling them to generate high-quality documents in seconds rather than hours. With a clean, intuitive dashboard and AI-driven capabilities, DocFlow AI represents a significant step forward in democratizing professional document creation.

Key Highlights

- ✓ AI-powered document generation using Claude Sonnet 4 model
- ✓ Resume analysis with ATS (Applicant Tracking System) scoring 0–100%
- ✓ 6 professional document templates: reports, proposals, API docs and more
- ✓ Real-time AI feedback and simulated HR professional perspective
- ✓ Secure user authentication with Streamlit session state management
- ✓ Clean, responsive dark-themed dashboard interface
- ✓ Modular architecture ready for future feature expansion

2. Project Overview

DocFlow AI is a full-stack web application built entirely in Python, leveraging the Streamlit framework for the frontend dashboard and the Anthropic Claude API for AI-powered content generation. The project was conceived to address the growing need for intelligent document automation tools.

Category	Details
Project Name	DocFlow AI
Type	AI-Powered Web Application / Dashboard
Framework	Streamlit (Python)
AI Integration	Anthropic Claude API — claude-sonnet-4-20250514
Language	Python 3.x
Architecture	Multi-page Streamlit App with session management
Deployment	Local server / Streamlit Community Cloud

Target Users	Students, Professionals, Researchers, Businesses
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3. Problem Statement & Purpose

The Problem

In today's fast-paced professional environment, creating high-quality documents remains time-consuming and tedious. Students spend hours writing internship reports. Professionals struggle with API documentation. Job seekers don't know if their resumes will pass ATS filters used by employers.

Time Waste	Creating professional documents from scratch takes 4–8 hours on average.
Quality Gap	Many users lack the writing skills to produce polished, professional documents.
ATS Blindness	Over 75% of resumes are rejected by ATS before a human ever reads them.
Template Scarcity	Good document templates are expensive or hard to customize for specific needs.
Tech Barrier	Non-technical users struggle with complex document formatting tools like LaTeX.

The Purpose

DocFlow AI was built to solve these exact problems. By harnessing the power of Claude AI, the platform enables anyone — regardless of writing ability or technical skill — to generate professional-grade documents in seconds. The mission is to democratize access to high-quality document creation and provide intelligent tools that give users a competitive edge in academic and professional settings.

4. Objectives

1

Primary Objective

Build a fully functional AI-powered document generation platform that is accessible, fast, and produces professional-quality output.

2

Secondary Objective

Integrate resume analysis with ATS scoring to help job seekers improve their chances of passing automated screening systems.

3

Technical Objective

Demonstrate practical application of Large Language Models in real-world productivity tools using Python and Streamlit.

4

UX Objective

Create an intuitive, visually appealing dashboard that requires no technical knowledge to operate.

5

Scalability Objective

Design a modular architecture allowing easy addition of new AI-powered modules such as PPT Generator and Content Assistant.

5. Features & Modules

■ Authentication System

Secure login and signup with Streamlit session state management. Includes Google OAuth placeholder and password reset flow.

- Email & password login
- New account registration
- Password reset via email link
- Session state management
- Auto-redirect after login

■ Dashboard Home

Central hub displaying key performance metrics, recent document history, and quick action buttons for all major features.

- 4 KPI metrics cards
- Recent documents list with type badges
- Quick action buttons

- Token usage progress bar
- User profile with plan info

■ AI Document Generator

4-step guided wizard for generating professional documents using Claude AI. Supports 6 templates with configurable parameters and download.

- 6 professional templates
- Internship Report, Project Report, Research Paper
- API Documentation, Business Proposal, Technical Specification
- Custom document option
- Step progress tracker
- Download as .txt

■ Resume Analyzer

Upload a PDF resume and paste a job description to receive comprehensive AI-powered ATS analysis with actionable recommendations.

- PDF resume upload & text extraction
- Job description matching
- ATS match score (0–100%)
- Strengths, improvements, keywords
- HR professional perspective simulation
- 4-tab results display

■ Settings

User profile management, AI model selection, and usage statistics with sign-out functionality.

- Profile editing (name, email)
- AI model selection (Claude / GPT-4o / Gemini)
- Token usage statistics
- Sign out

6. Technology Stack

Technology	Version	Purpose / Role
Python	3.x	Core programming language
Streamlit	1.57+	Web framework and dashboard UI
Anthropic Claude API	claude-sonnet-4	AI content & analysis engine
anthropic (SDK)	Latest	Python SDK for Claude API calls
PyPDF2	Latest	PDF text extraction for resume analysis
python-dotenv	Latest	Environment variable management (.env)
reportlab	4.x	PDF report generation
React / JSX	18.x	Original frontend prototype (converted)
IBM Plex Mono	Font	Monospace typography for UI

7. System Architecture

DocFlow AI follows a modular, layered architecture organized into three primary layers: the Presentation Layer (Streamlit UI), the Business Logic Layer (page modules), and the Service Layer (AI utilities).

File Structure

File / Folder	Description
app.py	Main entry point — auth routing and session state
pages/home.py	Dashboard shell, sidebar navigation, home metrics
pages/doc_generator.py	4-step AI document generation wizard
pages/resume_analyzer.py	PDF resume upload and AI analysis module
pages/settings.py	User profile and application settings page
utils/ai.py	Anthropic API wrapper — call_ai() utility function
.env	Environment variables (API keys, secrets)
.streamlit/config.toml	Streamlit config — hides auto-generated nav
venv/	Python virtual environment (isolated packages)

8. How It Works

Data Flow — Step by Step

1	Browser → app.py — User opens browser, app.py loads, checks <code>st.session_state.logged_in</code>
2	Authentication — If not logged in → Login form displayed → user enters credentials
3	Session Created — On successful login → <code>session_state</code> updated → <code>home.py</code> loaded
4	Navigation — User selects module from WORKSPACE sidebar → page module called via <code>show()</code>
5	Input Collection — User fills in form fields → data collected → AI prompt assembled
6	AI API Call — <code>call_ai()</code> sends structured prompt to Anthropic API → Claude generates response
7	Output Display — Response displayed in Streamlit UI → user can download or copy output

Document Generation — 4 Step Wizard

Step 1: Template: User selects from 6 predefined templates or starts from scratch.

Step 2: Configure: User fills title, company, duration, and additional context details.

Step 3: Generate: Parameters assembled into a prompt → sent to Claude API → document generated.

Step 4: Export: Document displayed with markdown rendering → downloadable as .txt file.

Resume Analysis Process

Upload	User uploads resume as PDF file via Streamlit file uploader.
Extraction	PyPDF2 extracts text content from all pages of the uploaded PDF.
Job Description	User pastes the target job description into a text area.
AI Analysis	Claude analyzes both texts and returns structured JSON with score, feedback.
Results	Results shown in 4 tabs: Strengths, Improvements, Keywords, HR View.

9. Use Cases & Target Users

Students

- Generate internship reports and project documentation automatically
- Create research papers with proper academic structure
- Analyze resume before applying for campus placements
- Write project reports for college assignments in minutes

Job Seekers

- Analyze resume against specific job descriptions
- Identify missing ATS keywords to improve visibility
- Get simulated HR professional feedback on resume quality
- Improve interview call rate with data-driven optimization

Software Developers

- Generate API documentation from scratch instantly
- Create technical specifications, SRS, and TDD documents
- Write business proposals and project plans for clients
- Produce system architecture and design documents

Business Professionals

- Create business proposals and executive summaries quickly
- Generate technical documentation for products and services
- Produce professional stakeholder reports and presentations
- Save 4–8 hours per document on average

10. Market Opportunity

The AI document automation market is experiencing explosive growth. The global AI writing assistant market is projected to reach \$6.5 billion by 2030, growing at a CAGR of over 25%. DocFlow AI is positioned at the intersection of several high-growth market segments.

Market Segment	Size / Growth	Relevance to DocFlow AI
AI Writing Tools	Growing 25%+ annually	Direct — Core product category
Resume Optimization	75M+ job seekers globally	Direct — Resume Analyzer module
Document Automation	\$6.5B market by 2030	Direct — Document Generator
EdTech (Students)	500M+ students globally	High — Primary user segment
HR Technology	\$30B+ global market	Medium — ATS analysis feature
Developer Tools	Millions of devs globally	Medium — API Documentation template

DocFlow AI's competitive advantage lies in its simplicity, Claude AI quality, and modular design. Unlike expensive enterprise tools, it provides professional-grade output accessible to anyone. The platform's open architecture allows rapid feature expansion to capture additional market share.

11. Future Enhancements

Sprint 2 — Coming Soon

- PPT Generator — AI-powered PowerPoint presentation creation
- Notes & Study AI — Smart note-taking and study material generation
- Content Assistant — Blog posts, social media content, marketing copy

Sprint 3 — Planned

- DOCX & PDF export — Direct download as formatted Word or PDF documents
- Multi-language support — Generate documents in multiple languages
- Template customization — User-defined templates and styles
- Collaboration features — Share and co-edit documents in real time

Sprint 4 — Vision

- Real user authentication with database backend (PostgreSQL / Firebase)
- Payment integration for premium Pro/Team plans
- Developer API access to integrate DocFlow AI into other products
- Mobile-responsive design and PWA support
- Advanced usage analytics dashboard for admins

12. Installation & Setup

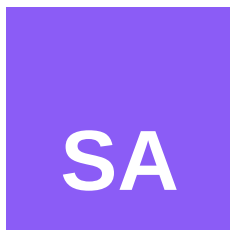
Prerequisites

- Python 3.8 or higher
- pip (Python package manager)
- Anthropic API key (from console.anthropic.com)
- Git (optional)

Step-by-Step Installation

1	Navigate to project	<code>cd C:\Users\your-name\Desktop\docflow-ai\docflow-streamlit</code>
2	Create virtual env	<code>python -m venv venv</code>
3	Activate venv	<code>venv\Scripts\activate</code>
4	Install packages	<code>pip install streamlit anthropic python-dotenv PyPDF2</code>
5	Add API key	<code>ANTHROPIC_API_KEY=your-api-key (in .env file)</code>
6	Run application	<code>streamlit run app.py</code>
7	Open in browser	<code>http://localhost:8501</code>

13. Creator Details



Full Name	Shuhaib Abdulla
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Role	Full Stack Developer / AI Enthusiast
Project	DocFlow AI — AI-Powered Document Platform
Tech Stack	Python, Streamlit, Anthropic Claude API, React JSX
Date	May 24, 2026

Shuhaib Abdulla is a passionate developer with expertise in Python, AI integration, and web application development. DocFlow AI represents a culmination of skills in full-stack development, AI API integration, and user experience design. The project demonstrates proficiency in building production-ready AI-powered tools that solve real-world problems for students, professionals, and businesses.

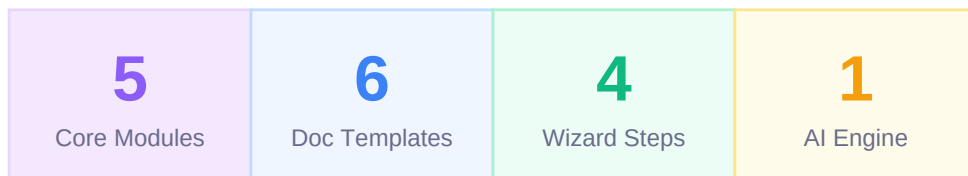
"Building tools that make AI accessible to everyone — regardless of technical background — is the driving philosophy behind DocFlow AI." — Shuhaib Abdulla

14. Conclusion

DocFlow AI represents a significant achievement in AI-powered application development. By successfully integrating Anthropic's Claude AI with a clean, accessible Streamlit dashboard, the platform delivers genuine value to students, professionals, and businesses who need high-quality documents quickly and efficiently.

The project demonstrates that AI tools don't need to be complex or expensive to be powerful. With a well-designed architecture, thoughtful user experience, and intelligent AI prompting, DocFlow AI delivers professional results that would otherwise take hours to produce manually.

The modular design ensures the platform can grow with user needs. With planned features including PPT generation, study notes AI, and content assistance, DocFlow AI is positioned to become a comprehensive AI productivity suite for the modern professional.



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